ModelIt!

Student Activity Pack

The Light You're Seeing Is Already Old

When You Look at a Star, You're Seeing the Past

Grade 5 | NGSS 5-ESS1-1 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think the sun looks so much brighter than other stars?

2 If light travels at 186,000 miles/second, and the nearest star is 4 light years away, how old is its light?

3 True or False: When you look at a distant star, you're seeing it as it is right now.

4 What is a light year?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Star Distance | Star Actual Luminosity | Light Travel Time | Apparent Brightness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: The Sun

Distance VERY LOW, Luminosity MODERATE

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Nearby Dim Star

Distance LOW, Luminosity LOW

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Distant Supergiant

Distance HIGH, Luminosity VERY HIGH

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Star Color / Temperature — Hotter stars are bluer and more luminous

Interstellar Dust — Dust clouds can block and dim starlight

Star Age — Stars change luminosity over their lifetime

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How do a star's distance and luminosity determine how bright it appears from Earth? Use evidence from your simulation to support your answers.

1. **Why does the sun appear so much brighter than other stars?**

2. **If a star is 1,000 light years away, how old is the light reaching us tonight?**

3. **What makes a star appear BRIGHTER even at great distance?**

4. **Could we be looking at a star that no longer exists? Explain.**

5. **What is the farthest thing human eyes can see without a telescope?**

8 STEM Challenge: Design a Star Observation Mission

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YOUR ENGINEERING MISSION

Design a mission to study a specific star using what you know about light travel time.

THE SCENARIO

NASA wants to study a star 100 light years away. What would observing it teach us? What are the challenges?

Engineering Design Thinking

1. WHAT STAR WILL YOU OBSERVE AND WHY?
2. WHAT INFORMATION CAN YOU GATHER FROM ITS LIGHT?
3. WHAT DOES LIGHT TRAVEL TIME MEAN FOR YOUR OBSERVATIONS?
4. HOW WOULD YOU COMMUNICATE FINDINGS TO THE PUBLIC?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Astronomers and Astrophysicists study everything beyond Earth using the principles you modeled today. They earn $90,000-$145,000/year.